Quantum Mechanics II Excercise 3

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1 $SU(2), SO(3)$

1.1 $SU(2)$ and the Pauli matrices

$SU(2)$ is defined as the set of all 2×2 complex matrices such that $U^\dagger U = \mathbb{1}$ and $\det U = 1$. We can characterise those matrices by starting from a generic 2×2 matrix and narrowing down using the conditions. Given a 2×2 matrix U, it can be written as

$$U = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \tag{1}$$

The first condition for it to be in $SU(2)$ is $U^\dagger U = \mathbb{1}$. This means that the matrix is invertible and that its inverse is its complex conjugate

$$U^\dagger = U^{-1} \tag{2}$$

Using the formula for the inverse of a 2×2 matrix:

$$U^{-1} = \frac{1}{|\det U|} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \tag{3}$$

And using the second condition $\det U = 1$ we get:

$$\begin{pmatrix} a^* & c^* \\ b^* & d^* \end{pmatrix} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \tag{4}$$

$$a^* = d \qquad d^* = a \qquad b^* = -c \qquad c^* = -b \tag{5}$$

In other words, we can write U as:

$$U = \begin{pmatrix} a & b \\ -b^* & a^* \end{pmatrix} \tag{6}$$

Writing a and b in their complex cartesian representation, we get:

$$U = \begin{pmatrix} x + iy & w + iz \\ -w + iz & x - iy \end{pmatrix} \tag{7}$$

And requiring $\det U = 1$ gives us:

$$x^2 + y^2 + w^2 + z^2 = 1 \tag{8}$$

Now, let us look at the expression

$$A = \exp\left[\frac{1}{2} i\theta_i \sigma_i\right] \tag{9}$$

Using the definition of the Pauli matrices, we get

$$A = \exp\left[\frac{1}{2} i\theta_i \sigma_i\right] = \exp\begin{pmatrix} i\theta_z & \theta_y + i\theta_x \\ -\theta_y + i\theta_x & -i\theta_z \end{pmatrix} \equiv \exp P(\theta) \tag{10}$$

Where we named the matrix in the exponent $P(\theta)$. We can now expand the exponent, but first let us calculate the powers of $P(\theta)$:

$$P^2(\theta) = \begin{pmatrix} i\theta_z & \theta_y + i\theta_x \\ -\theta_y + i\theta_x & -i\theta_z \end{pmatrix} \begin{pmatrix} i\theta_z & \theta_y + i\theta_x \\ -\theta_y + i\theta_x & -i\theta_z \end{pmatrix} = -\begin{pmatrix} \theta_x^2 + \theta_y^2 + \theta_z^2 & 0 \\ 0 & \theta_x^2 + \theta_y^2 + \theta_z^2 \end{pmatrix} \equiv -\theta^2 \mathbb{1} \tag{11}$$

Therefore, we can split the powers of P to even and odd powers:

$$P^{2n}(\theta) = (-1)^n \, \theta^{2n} \mathbb{1} \qquad P^{2n+1}(\theta) = (-1)^n \, \theta^{2n} P \tag{12}$$

And thus

$$\begin{aligned} A = \exp P = \sum_{n=0}^{\infty} \frac{P^n}{n!} = \mathbb{1} \sum_{n \ even} \frac{(i\theta)^n}{n!} + P \sum_{n \ odd} \frac{(i\theta)^{n-1}}{n!} = \cos(\theta)\mathbb{1} + \sin(\theta)\frac{P}{\theta} = \\ = \begin{pmatrix} \cos(\theta) + i\sin(\theta)\frac{\theta_z}{\theta} & \sin(\theta)\frac{\theta_y + i\theta_x}{\theta} \\ \sin(\theta)\frac{-\theta_y + i\theta_x}{\theta} & \cos(\theta) - i\sin(\theta)\frac{\theta_z}{\theta} \end{pmatrix} \end{aligned} \tag{13}$$

The determinant of this matrix is:

$$\begin{aligned} \det A = \left(\cos(\theta) + i\sin(\theta)\frac{\theta_z}{\theta}\right)\left(\cos(\theta) - i\sin(\theta)\frac{\theta_z}{\theta}\right) - \sin^2(\theta)\frac{-\theta_x^2 - \theta_y^2}{\theta^2} = \\ = \cos^2(\theta) + \sin^2(\theta)\frac{\theta_z^2}{\theta^2} + \sin^2(\theta)\frac{\theta_x^2 + \theta_y^2}{\theta^2} = \cos^2(\theta) + \sin^2(\theta)\frac{\theta_x^2 + \theta_y^2 + \theta_z^2}{\theta^2} = \\ = \cos^2(\theta) + \sin^2(\theta) = 1 \end{aligned} \tag{14}$$

Which means that with a small change of variables we can make it fit the matrix in equation 7:

$$x = \cos(\theta) \qquad y = \sin(\theta)\frac{\theta_z}{\theta} \qquad w = \sin(\theta)\frac{\theta_y}{\theta} \qquad z = \sin(\theta)\frac{\theta_x}{\theta} \tag{15}$$

Which means that for every 2×2 matrix U in $SU(2)$ one can find $\theta_x, \theta_y, \theta_z$ such that $U = \exp\left[\frac{1}{2} i\theta_i\sigma_i\right]$, and thus $t_i = \frac{1}{2}\sigma_i$ furnish a representation of the $su(2)$ algebra.

1.2 The $SO(3)$ group

For the $SO(3)$ group, we can follow the same pattern. Starting from a 3×3 matrix

$$O = \begin{pmatrix} a & b & c \\ d & e & f \\ h & k & l \end{pmatrix} \tag{16}$$

And demanding that $O^T O = \mathbb{1}$, we get:

$$O^T = \begin{pmatrix} a & d & h \\ b & e & k \\ c & f & l \end{pmatrix} \tag{17}$$

$$O^T O = \begin{pmatrix} a^2 + d^2 + h^2 & ab + de + hk & ac + df + hl \\ ab + de + hk & b^2 + e^2 + k^2 & bc + ef + kl \\ ac + df + hl & bc + ef + kl & c^2 + f^2 + l^2 \end{pmatrix} \tag{18}$$

Which gives the equations

$$a^2 + d^2 + h^2 = b^2 + e^2 + k^2 = c^2 + f^2 + l^2 = 1 \tag{19}$$

$$ab + de + hk = ac + df + hl = bc + ef + kl = 0 \tag{20}$$

The three matrices that make up the representation of the algebra are:

$$L_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix} \qquad L_2 = \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix} \qquad L_3 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{21}$$

Their linear sum with constant φ_i is

$$P = (-i)i \begin{pmatrix} 0 & -\varphi_3 & \varphi_2 \\ \varphi_3 & 0 & -\varphi_1 \\ -\varphi_2 & \varphi_1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -\varphi_3 & \varphi_2 \\ \varphi_3 & 0 & -\varphi_1 \\ -\varphi_2 & \varphi_1 & 0 \end{pmatrix} \tag{22}$$

Which means that

$$A = \exp\left[-i\varphi_i L_i\right] = \exp P = \exp \begin{pmatrix} 0 & -\varphi_3 & \varphi_2 \\ \varphi_3 & 0 & -\varphi_1 \\ -\varphi_2 & \varphi_1 & 0 \end{pmatrix} \tag{23}$$

We can now look at the powers of P to get

$$P^2 = \begin{pmatrix} 0 & -\varphi_3 & \varphi_2 \\ \varphi_3 & 0 & -\varphi_1 \\ -\varphi_2 & \varphi_1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -\varphi_3 & \varphi_2 \\ \varphi_3 & 0 & -\varphi_1 \\ -\varphi_2 & \varphi_1 & 0 \end{pmatrix} = \begin{pmatrix} -\varphi_2^2 - \varphi_3^2 & \varphi_1\varphi_2 & \varphi_1\varphi_3 \\ \varphi_1\varphi_2 & -\varphi_1^2 - \varphi_3^2 & \varphi_2\varphi_3 \\ \varphi_1\varphi_3 & \varphi_2\varphi_3 & -\varphi_1^2 - \varphi_2^2 \end{pmatrix} \tag{24}$$

$$P^3 = \begin{pmatrix} 0 & -\varphi_3 & \varphi_2 \\ \varphi_3 & 0 & -\varphi_1 \\ -\varphi_2 & \varphi_1 & 0 \end{pmatrix} \begin{pmatrix} -\varphi_2^2 - \varphi_3^2 & \varphi_1\varphi_2 & \varphi_1\varphi_3 \\ \varphi_1\varphi_2 & -\varphi_1^2 - \varphi_3^2 & \varphi_2\varphi_3 \\ \varphi_1\varphi_3 & \varphi_2\varphi_3 & -\varphi_1^2 - \varphi_2^2 \end{pmatrix} = -\varphi^2 P \tag{25}$$

Then, for $n > 0$

$$P^{2n} = (i\varphi)^{2n-2} P^2 \qquad P^{2n+1} = (i\varphi)^{2n} P \tag{26}$$

$$\begin{aligned} A = \exp P = \sum_{n=0}^{\infty} \frac{P^n}{n!} = \sum_{n\ even} \frac{P^n}{n!} + \sum_{n\ odd} \frac{P^n}{n!} = \mathbb{1} + \sum_{n>0\ even} \frac{(i\varphi)^{n-1}}{n!} P^2 + \sum_{n\ odd} \frac{(i\varphi)^n}{n!} P = \\ = \mathbb{1} - i\frac{\cos(\varphi) - 1}{\varphi} P^2 + \sin(\varphi) P \end{aligned} \tag{27}$$

1.3 Commutation relations

It's easy to see that for $t_i = \frac{1}{2}\sigma_i$, $[t_i, t_j] = i\varepsilon_{ijk} t_k$. The simplest way is just to check them one by one:

$$\begin{gathered} [t_1, t_1] = 0 \qquad [t_2, t_2] = 0 \qquad [t_3, t_3] = 0 \\ [t_1, t_2] = -[t_2, t_1] = t_1 t_2 - t_2 t_1 = \\ = \frac{1}{4}\left[\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} - \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\right] = \frac{1}{4}\left[\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} - \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix}\right] = \\ = \frac{1}{4}\left[i\sigma_3 - (-i)\,\sigma_3\right] = \frac{1}{2} i\sigma_3 = it_3 \\ [t_1, t_3] = -[t_3, t_1] = t_1 t_3 - t_3 t_1 = \\ = \frac{1}{4}\left[\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\right] = \frac{1}{4}\left[\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\right] = \\ = \frac{1}{4}\left[-i\sigma_2 + (-i)\,\sigma_2\right] = -\frac{1}{2} i\sigma_2 = -it_2 \\ [t_2, t_3] = -[t_3, t_2] = t_2 t_3 - t_3 t_2 = \\ = \frac{1}{4}\left[\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\right] = \frac{1}{4}\left[\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} - \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}\right] = \\ = \frac{1}{4}\left[i\sigma_1 - (-i)\,\sigma_1\right] = \frac{1}{2} i\sigma_1 = it_1 \end{gathered} \tag{28}$$

Similarly, for Li we get

$$
\begin{gathered}
[L_1, L_1] = 0 \qquad [L_2, L_2] = 0 \qquad [L_3, L_3] = 0 \\
[L_1, L_2] = -[L_2, L_1] = L_1 L_2 - L_2 L_1 = \\
= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix} = \\
= \begin{pmatrix} 0 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = i \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = iL_3 \\
[L_1, L_3] = -[L_3, L_1] = L_1 L_3 - L_3 L_1 = \\
= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix} = \\
= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = i \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix} = -iL_2 \\
[L_2, L_3] = -[L_3, L_2] = L_2 L_3 - L_3 L_2 = \\
= \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix} = \\
= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & -1 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix} = i \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix} = iL_1
\end{gathered} \tag{29}
$$

This means that both t_i and L_i satisfy the same commutation relations, and there is an obvious isomorphism between their respective algebras.

2 A bit of $SO(1,3)$

3 What adds under Lorentz transformations?